5 Magnetic field control

The magnetic field control system consists of the magnetic shield, the set of shim coils, and the internal and external magnetometers. Together, these components allow for interaction region operation with low levels of ambient magnetic background present. This section will discuss the design requirements of the system, present the physical details of its implementation, as well as the performance achieved with the present shielding system.

While the design, building, and characterization of the shield can be done in a straightforward manner by emulating past work, the underlying science is frequently not obvious and is often left out of the literature on the topic. This is understandable, since understanding how the shield works is rarely of interest to the experimentalist desirous of getting an atomic or molecular beam operational. In what follows, I will also have to leave several lines of exploration unfinished as they are not immediately required in order to proceed with the CENTREX experiment. However, it may be a starting point for future work requiring higher performance.

5.1 Design requirements

5.2 Theory and related work

5.3 Physical structure

The magnetic shield consists of four concentric cylindrical layers with open ends (See Figs. 17 and 18). Three of these are built by wrapping metglas foil around PVC pipes, while the fourth, innermost layer has been welded out of two sheets of mu-metal rolled into a tubular form, and annealed. The mu-metal tube, being 1 mm thick, is relatively fragile; in order to limit its motion, it is sandwiched between the hard plastic PVC tubes of the inner metglas layer and that of the shim coils. The inner two metglas layers are spaced apart using small-diameter PVC tubing; the same method is employed to suspend the glass vacuum chamber of the interaction region centered inside the shim coils tube. The outermost tube is mounted on an 80/20 frame using a set of seven aluminum semi-circular plates.

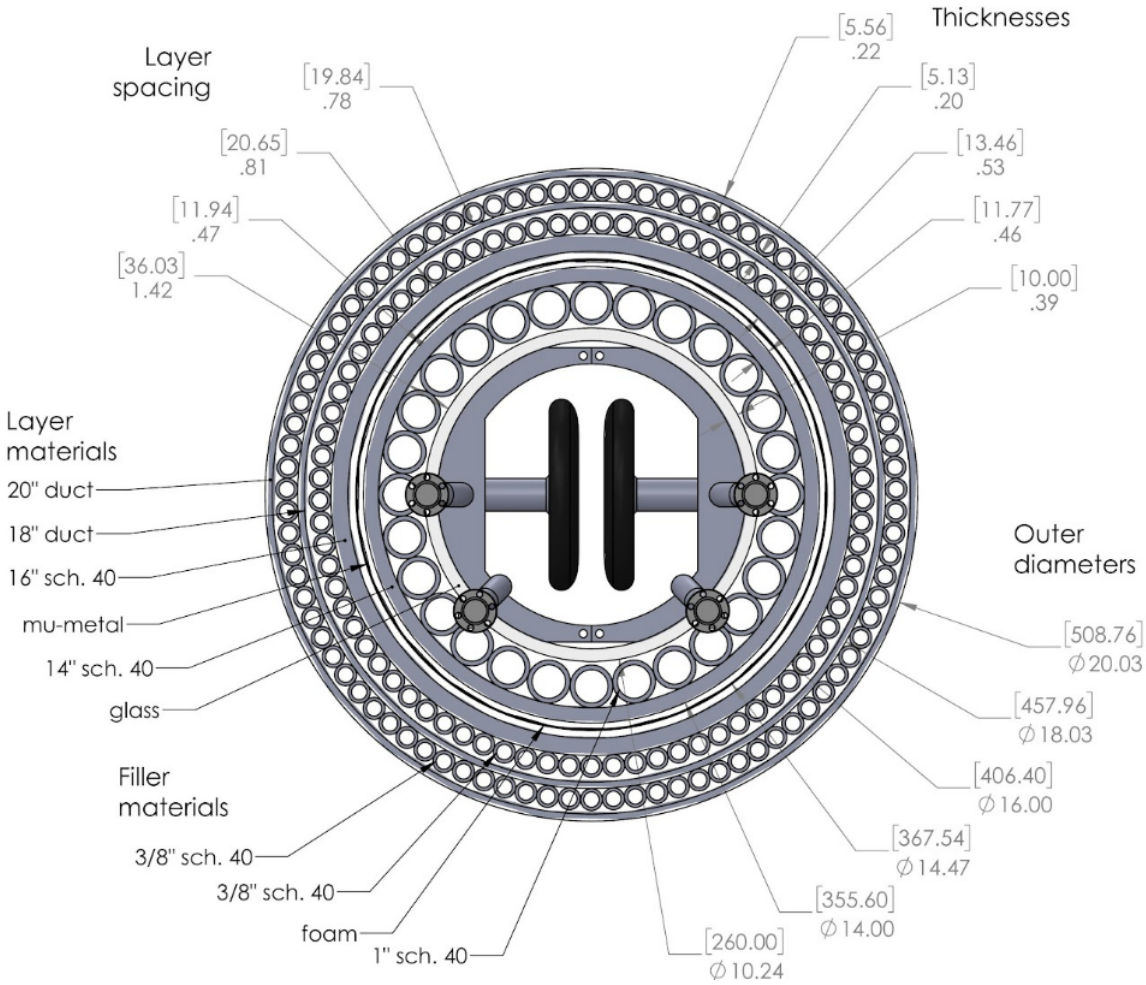

Figure 17: Cross-sectional view of the magnetic shield.

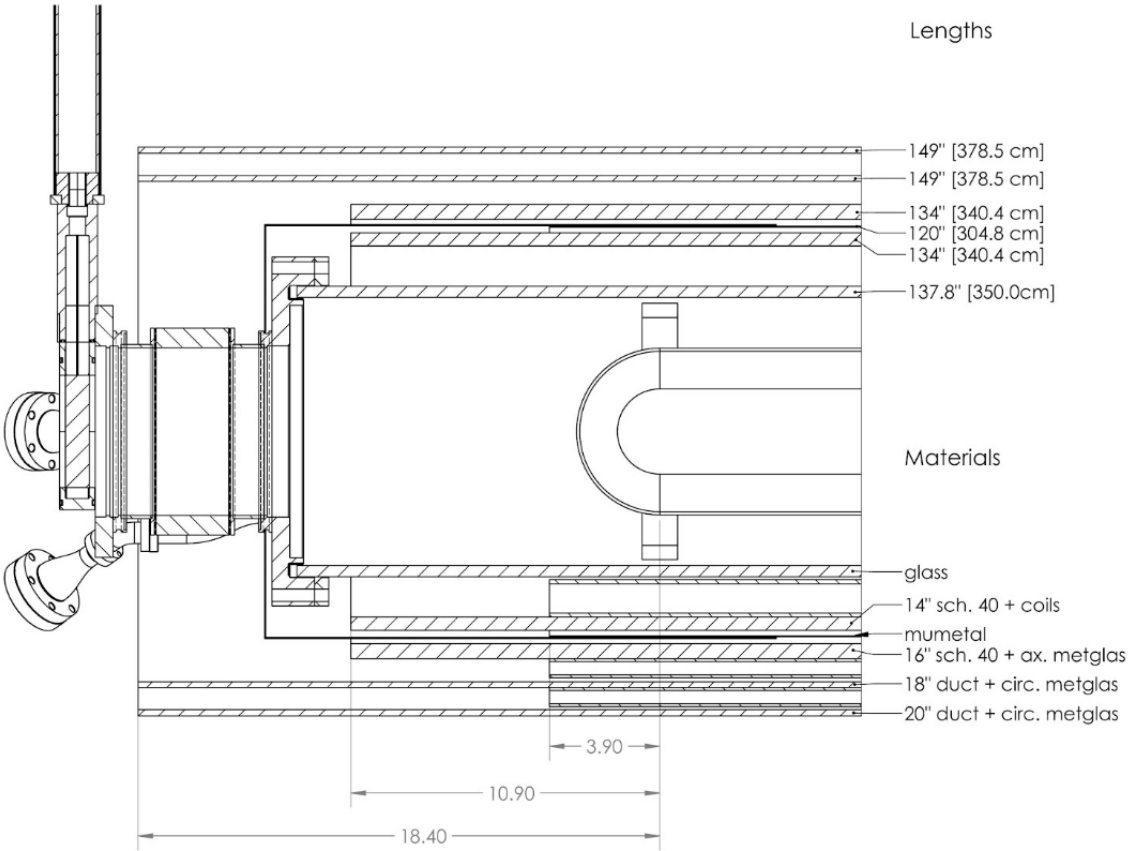

Figure 18: Longitudinal cross-section of the magnetic shield, showing the unequal lengths of the layers.

Layers, spacers, mounting The three metglas layers are all built in the same way. First, the plastic tube (provided by the Harrison Machine & Plastic Corp., Garrettsville, OH) is cut to size with a handsaw if necessary (see Table V for layer lengths and material details) and roughly cleaned with water and paper towels on the outside and, as far as possible, also on the inside to remove the dirt it comes with, probably due to outdoors storage. Then, I cut 8×12 pieces of metglas (7×12 for the 16" layer) to the length required to wrap around the tube with approx. 1" of overlap. The pieces are secured to the PVC tube and to each other with a few short pieces of Kapton tape (McMaster p.n. 7648A63) so that they do not fall down during assembly.

The prepared and cleaned tubes are put on a stand with wheels under them so they can be rotated, facilitating the circumferential wrapping of the metglas foil. The first piece of tape is wrapped about an inch from the end of the tube; leaving this much bare PVC

Table V: The four layers of the magnetic shield.

Layer	Length [in]	Weight [lbs]
Main layers:		
20” duct	149	100
18” duct	149	80
16” sch. 40	134	180
14” sch. 40	134	140
Spacers:		
3/8” sch. 40	120*	190
1” sch. 40	120*	90

helps with handling the approx. 100–200 lb tubes. The second piece is wrapped so as to have about an inch of overlap with the first, and the remaining six pieces are attached analogously. Once the first layer is done, the wrapping resumes at the position of the first metglas piece. This procedure is repeated for a total of 12 layers total. (Due to axial and circumferential overlaps, the true thickness of the shield is not perfectly homogeneous, but in practice we find that this is unimportant as long as the overall shielding is sufficiently high.)

To secure the metglas pieces and prevent them from coming loose in case the Kapton tape fails, the entire tube is wrapped in plastic foil. Following that, there are 12 turns of degaussing wire wrapped in a toroidal fashion around the tube. The wire is 22 or 24 gauge lacquered copper wire, sufficiently thick for approx. 2 amps of degaussing current. The 12 turns are uniformly distributed around the circumference of the tube; we believe this may be important for the performance of the degaussing. On the outermost (20”) tube only, we also have a solenoidal coil with approx 1 turn per in spacing for rough cancellation of axial fields. (Given the metglas wrapping geometry, this solenoidal coil would not be effective for degaussing.) Finally, each tube is again wrapped in plastic wrap to secure the degaussing and (for 20” tube only) the solenoidal coil.

The mu-metal layer does not, of course, require wrapping with metglas, only the degaussing coil and the outer plastic foil wrap. However, care must be taken to reinforce the insulation of the degaussing wire at the ends of the cylinder; the sharp edges can easily strip the magnet wire coating and short the turns of the wire to the conductive cylinder,

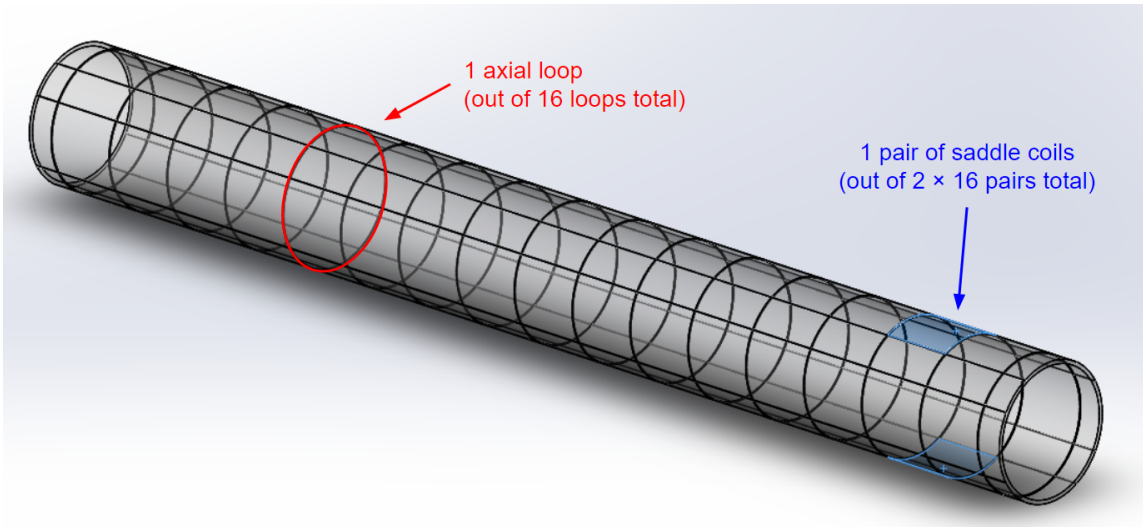

Figure 19: Shim coil geometry.

which does not happen with the PVC tubes. I used heat shrink tubing to do this, but it required cutting the wire at each turn and at both ends, adding insulation, soldering the wires back together and heating the insulation to make it shrink and adhere. It would have been better to use a wire with more durable insulation instead.

Shim coils The set of shim coils consists of 16 segments of three dimensions each (see Fig. 19). The two transverse dimensions (B_x and B_y) are implemented as pairs of one-loop saddle coils (8" by 60°each), while the axial field is controlled by simple circular current loops (spaced 8" apart from each other). This gives a total of $5 \times 16 = 80$ degrees of freedom. The 22g magnet wires are fixed in place by being glued into a series of grooves on the surface of the 14" PVC tube using neutral-cure RTV silicone sealant (Dow Dowsil 737) and wrapped in plastic foil for added protection. During the curing of the silicone, the wires are held in place by Kapton tape, which is removed after the silicone hardens.

The size of the tube makes it impractical to use standard milling machines. While the circular grooves can easily be cut using a hand saw, the axial ones were done on a custom-made setup (Fig. 20) involving a hand drill clamped on a 80/20 frame, which in turn is mounted using linear bearings to a larger frame. The setup is not as rigid as a true milling machine would be, which is why it is important to use a constant force while moving the carriage forward to avoid cutting wavy grooves. I used a system of ropes, pulleys, and

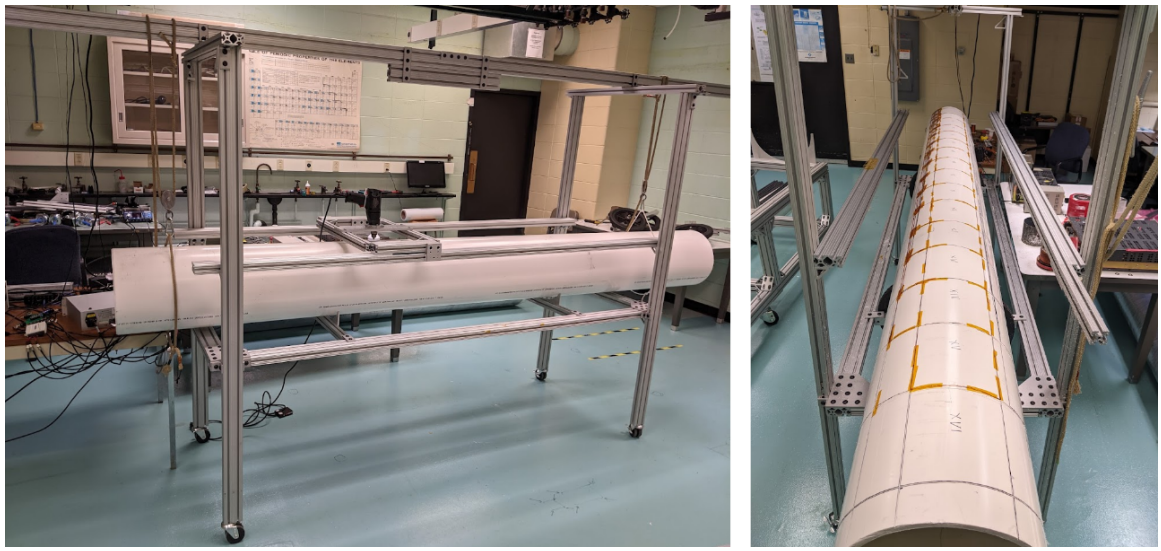

Figure 20: ***a.*** Milling machine for cutting longitudinal grooves. ***b.*** Wires taped inside grooves while silicone hardens.

weights to achieve satisfactory results.

With 80 different pair of wires (twisted into twisted pairs using a hand drill), cable management is essential so as not to confuse the cables with each other. For each coil, a linear groove is cut from one end of the tube to the location of the coil; thus all coils can be connected to at one end of the shield. There, the sets of wire pairs belonging to each dimension are bundled together into five bundles of 16 pairs using zip ties; each wire pair is temporarily labelled by a numbered piece of Kapton tape at the end. For ease of connecting and disconnecting, these wires are attached to adapter boards that are permanently attached to the 80/20 support frame of the shield. The adaptor boards bundle four pairs of wires into one RJ45 jack to enable using standard patch cables for ease of connection.

Support structure The considerable total weight of the magnetic shield (780 lbs of plastic, plus metglas, mu-metal, and other materials) requires a sufficiently strong support structure. In our case, it is assembled from 80/20 rails (profile number 3030) and placed on eight caster wheels McM p.n. 9813T71) so the setup can in principle be moved by one person. The support structure consists of three four-foot segments mounted together to form a rectangular box. Rails going across the top of the box hold aluminum plates (1/4" thick) with semi-circular cutouts. To keep the plates from damaging the delicate outer

metglas layer, the sharp corners of the cutouts are protected with hard nylon tape.

5.4 Operation and measurements

The characterization and use of the shield is an iterative process requiring a repeated sequence of three operations: magnetic field measurement, degaussing, and shimming.

I distinguish two types of measurement of the magnetic field of interest to us. Residual field is the field inside the shield assuming no changes in the external magnetic conditions. In our four-layer shield, this is the dominant magnetic field inside the interaction region and it is driven primarily by the interaction between the magnetization of the ferromagnetic shield layers with the external magnetic field. Despite our best attempts at degaussing, these residual fields are typically of the order of 50–200 μG in the center of the shield.

Field measurement During shield characterization, the primary measurement is of the field in the central region of the shield. During the use of the shield in a beam-line experiment, this region is difficult to access; it is also the region of greatest interest since all molecules included in the experiment pass through it. During characterization I attached a 4" $\times$ 4" aluminium rectangular tube by suspending it from them shield mounting structure at each end. I used a system of pulleys to pull the sensor carriage using a stepper motor and a rotary encoder to measure the position of the sensor along the length of tube. The sensor itself (Bartington Mag03MS-100) is a 3-channel fluxgate magnetometer with a measuring range of 1 G and a typical electronic offset of several hundred microgauss. (The specified noise level of 0.1 $\mu\mathrm{G}/\sqrt{\mathrm{Hz}}$ at 1 Hz, integrated over the 3 kHz bandwidth of the sensor, is below the mechanical stability of the sensor translation setup with a typical repeatability of approx 2 μG.)

The electronic offset of the sensor and its associated electronics is significant. It is specified as 50 μG, with a temperature coefficient of 1 μG/°C, although in practice offsets of several hundred microgauss are typical. The offset calibration is done by taking a measurement and immediately repeating it with the sensor rotated upside down. This is accomplished by mounting the sensor in a short rectangular aluminum tube with enough plastic tape on the outside so that the sensor carriage fits tightly inside the long square tube

going through the center of the shield. After a measurement is done, the sensor is pulled out of the long tube and reinserted rotated upside down. The accuracy of the offset calibration then depends on how well-centered the sensor is in the carriage, how much the field inside the shield changes during the time it takes to perform the calibration, how much the electronic offset itself changes in the same time, and how perfectly square the cross-sections of the long tube and the carriage are.

Rather than attempt to estimate the magnitudes of these effects individually, I estimate the precision/accuracy of the calibration from the calibration curve itself; see Fig. 21 for a typical example. It shows the two transverse components of the magnetic field (B_x and B_y) as measured by the sensor in its normal orientation, as well as with the sensor inverted. The data is presented in the symmetrized form: (normal + inverted)/2, which corresponds to the sensor offset, and (normal − inverted)/2, which corresponds to the physical magnetic field plus any calibration imperfections. Given the time scale of the changes in the electronic offset is primarily driven by temperature changes, which are negligible in the time it takes to do the calibration (around two minutes), we expect the curves showing the offset to be perfectly flat within the noise of the measurement, but that is typically not the case. The RMS deviations from the average value I identify as the error in the offset calibration, consisting of all the aforementioned factors.

While one may take the (normal − inverted)/2 superposition as the best measurement of the physical fields, doing so means that the error is different at each location along the shield. Since we expect the offset to be essentially constant during the measurement, I believe it a better practice to estimate the offset from the average value of the superposition, and then simply subtract this constant value from all subsequent measurements. This way, the overall offset is subject to an uncertainty, but the shape of the field pattern is known much more precisely.

The typical repeatability of a measurement can be within ± 1 μG, as shown in the example measurement in Fig. 22. In addition to the random errors introduced by the mechanical instability of the measurement setup, we also have to pay attention to the systematic errors induced by the signal chain from the sensor onwards. The sensor and its preamplifier have a limited bandwidth; 3 kHz for the sensor, and a settable low-pass filter for

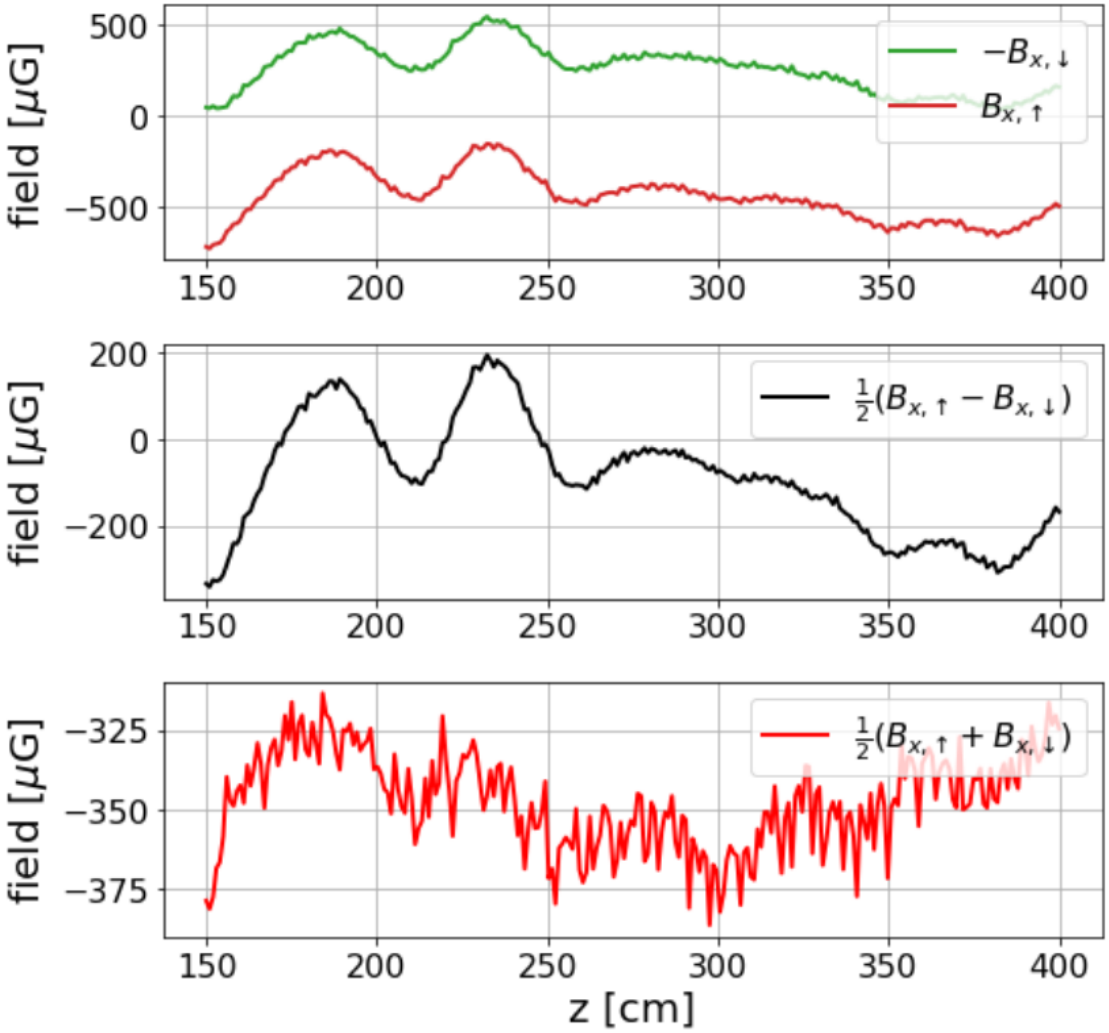

Figure 21: Electronic offset of the magnetometer as a function of position along the length of the center of the magnetic shield. Only the two transverse dimensions are shown.

the preamp. This must not be set too low in relation to the speed of the sensor motion. In practice, I use the lowest reasonable LPF cutoff frequency, verifying the choice by repeating the measurement with the next higher available frequency. If the two measurements are sufficiently close to each other, I use the lower of the two frequencies. I use a similar process to decide upon the digital LPF included with the ADC.

Dynamic field measurement The attenuation of the external fields is not perfect. If an infinitely large Helmholtz pair were to drive a perfectly uniform and sinusoidally varying magnetic field in the region of space occupied by the magnetic shield, then a smaller sinusoidal variation of the field inside the shield could be detected that is of the same frequency as the external field. In practice, there are practical limits to the size of exteral coils. In one version of the measurement, I have attached a room-sized loop of wire to the walls of the lab and drive them with a function generator while passing the magnetometer reading through a lock-in amplifier. Repeating the measurement in the absence of any shielding gives the magitude of the external field variation. Dividing this by the magnitude measured with the shield in place gives the shielding factor.

The disadvantage of this method is that the magnetic shield has to be removed to determine the external magnetic field, since the shape of the external coils is difficult to measure precisely. To find the SF of the final shield, I have used smaller rectangular coils of well-known dimensions centered around the magnetic shield; the field they apply, while not uniform, can be calculated using a closed-form expression.

Degaussing Degaussing is a method of reducing the residual field inside a magnetic shield based on the following observation: the field inside a hollow cylinder made of a ferromagnetic material, when placed in a region of a given external field, can be greatly reduced by driving a sinusoidal magnetic field with a gradually decreasing envelope.

The hardware used for degaussing in this experiment is as follows. Each of the four shielding layers has its own 12-turn degaussing coil. These are connected to a PCB with a set of relays that can connect each of them to the degaussing current, but only one at a time. During degaussing, the outer layer is always connected first, followed by the

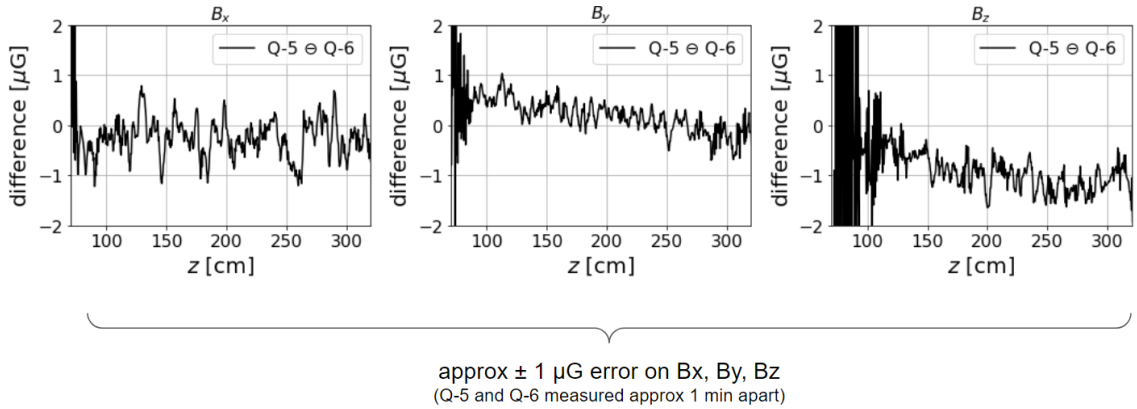

Figure 22: The repeatability of the measurement, shown as the difference between two measurements, one taken immediately after the other.

second outermost layer, followed by the innermost metglas layer, followed by the mu-metal layer. Since the residual field is primarily determined by the innermost layer, that one is degaussed last lest it be magnetized by accident during the degaussing of the other layers. The relays PCB is in turn connected to an isolation transformer for the purpose of DC blocking. We have considered alternate forms of DC blocking, such as a capacitor, but the required capacitance was too high at the low frequencies typically used in degaussing (down to 1 Hz). The transformer is driven by a power amplifier (Kepco BOP36-12M), which in turn is driven by a home-built DAC board. figure of connections

A wide rage of degaussing waveforms have been used in the literature; for example, linear or polynomial envelopes [REF] or envelopes specifically optimized for a linear decrease in the magnetization of the ferromagnetic material [REF]. In this experiment, I use a finite number of sinusoidal degaussing pulses. The pulses can optionally be separated by and adjustable delay to probe the evolution of the residual field from pulse to pulse. Each subsequent pulse has its amplitude decreased by a constant factor compared with the preceding pulse; thus our protocol is similar to, but not identical to, using an exponentially decreasing envelope multiplying a sine wave. fig: parameters of the degaussing waveform

Despite spending significant amounts of time on optimizing the degaussing parameters, I am only able to claim the following. The degaussing frequency must be slow enough that the skin effect does not significantly affect the degaussing; the initial amplitude has to be large enough to drive the shield into saturation; the final amplitude has to be small enough

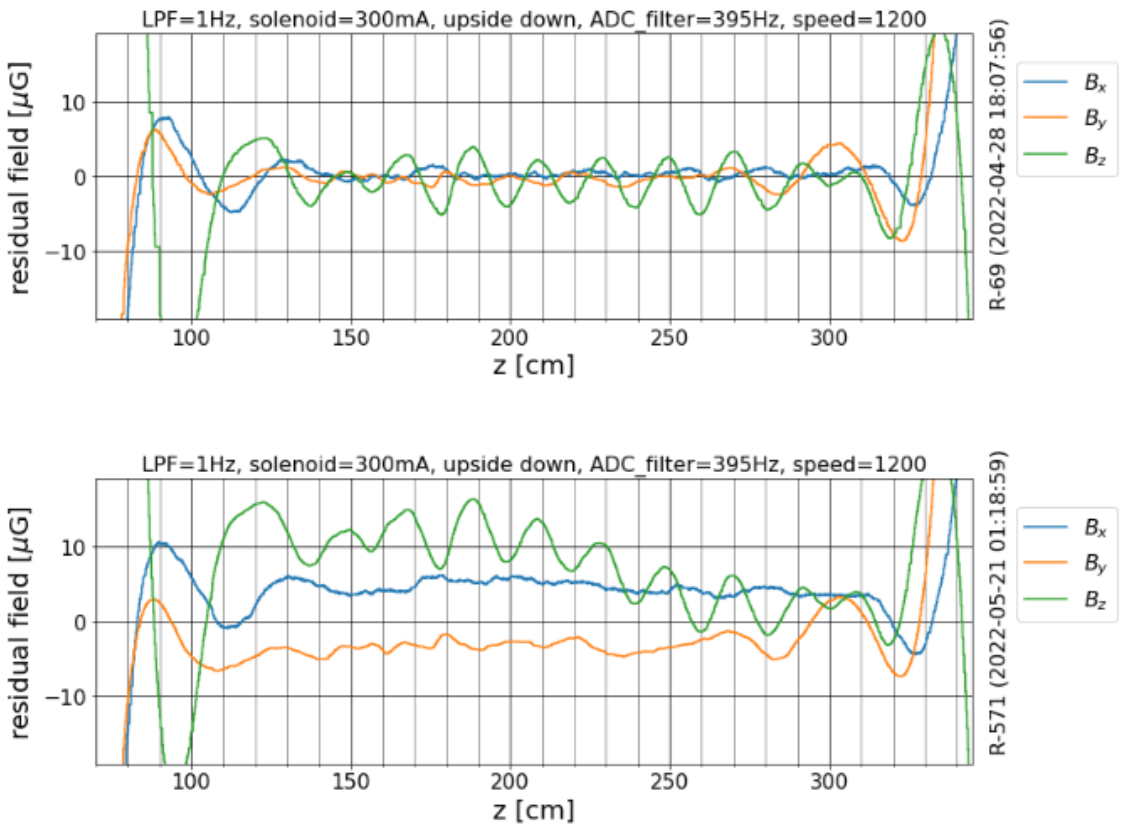

Figure 23: The magnetic field ***a.*** immediately after shimming, and ***b.*** three weeks later.

that the shield has recovered from saturation and is circling near a point around the origin of the hysteresis curve; the number of degaussing pulses has to be large enough so that the decrease in amplitude is gradual. The exact values of the parameters mentioned in the preceding statements need to be found experimentally for the shield geometry used; fig: degaussing optimization.

Exploration of these parameters is further complicated by the fact that there are four layers to degauss and degaussing one may inadvertently magnetize the others. I have tried degaussing several or all layers in parallel, but have not found any significant improvement and thus have not further explored that method. However, while building the individual metglas layers, I have measured the magnetization of each layer before and after degaussing; fig: data on individual layers before/after degaussing

Given the importance of the long-term stability of the magnetic fields inside the shield, it is interesting to explore how stable the degaussing process is, i.e., how much does the

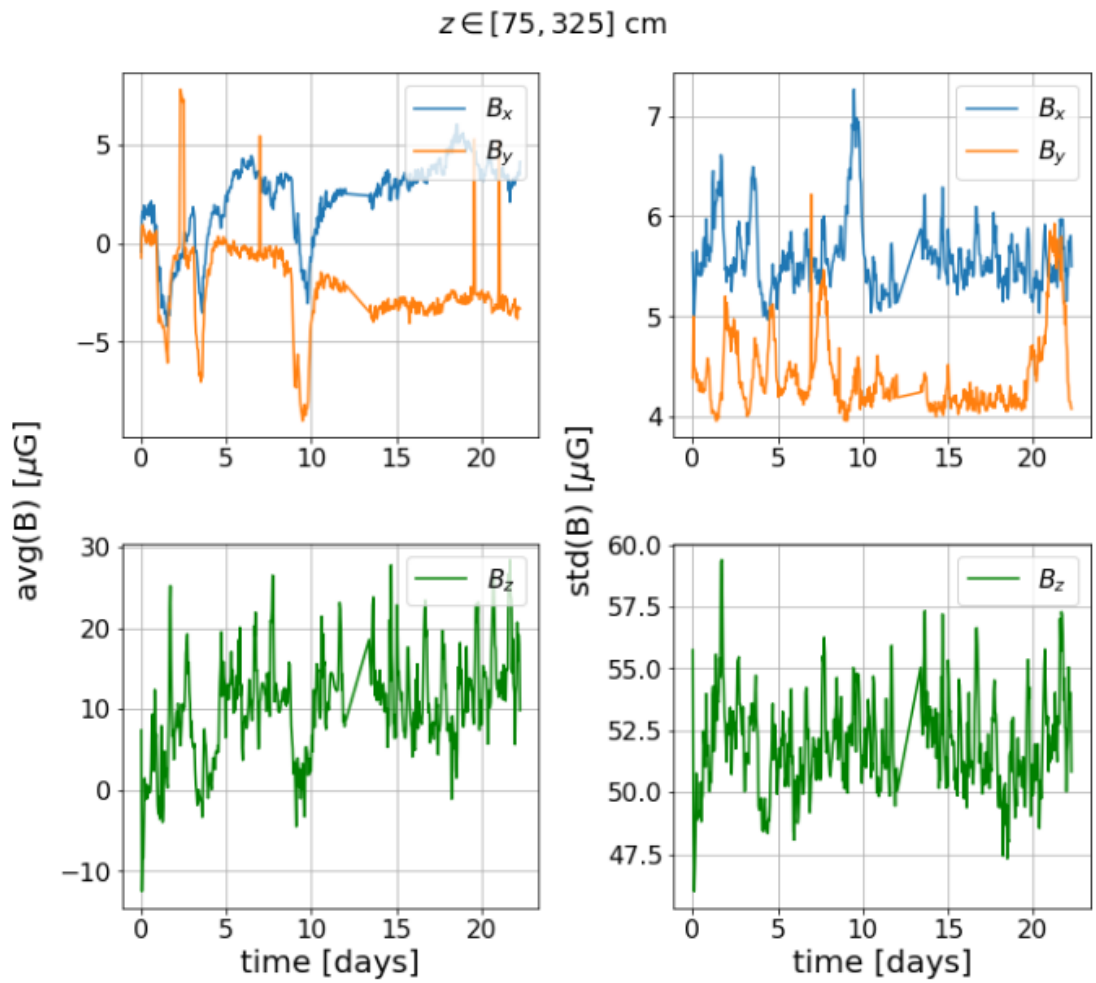

Figure 24: Average and RMS magnetic field in the central 2.5 m of the shield over the course of three weeks.

field change when degaussing repeatedly, or what happens when the field has drifted away from the degaussed value and the shield is degaussed again. figure

Shim coils Once the shield has been degaussed as well as is possible, the remaining magnetic field has to be cancelled using the system of shim coils. The accuracy of this depends on the measurement of the residual field, as well as on the geometry of the coils themselves. During shield characterization, we have access to the center of the shield and can measure the fields to within 1–2 μG. Thus the limit is set by the number of coil segments and their size.

The shimming process consists of two steps. First, the individual field patterns generated

by each of the 80 shim coils have to be measured. I do this by flowing a constant current through the coil under test, measuring the magnetic field, then reversing the direction of the current and repeating the measurement. By subtracting the two measurements, applying smoothing, and masking the parts of the curve where the coil field is expected to be negligible fig: example calibration curve. Once the calibration curves for all the coils are known, I use a simple least-squares minimization routine to determine the optimal currents to use so as to minimize the magnitude of the residual field inside the shield. While I could limit the optimization routine to only minimize the field along the vertical axis, this being the component we are most sensitive to in the molecular beam experiment, I do not anticipate a significant improvement in the shimming by doing so given that the three dimensions of the magnetic field accessed by the coils are orthogonal to each other.

5.5 Performance and limitations

The design requirements for the shield specified a 10 μG maximum average residual field. This has been met; see Fig. 23 for a typical measurement of the fields inside after shimming.

Of particular interest is how the fields develop after the shimming procedure. To this end, I have followed their evolution over the course of three weeks, finding that neither the average nor the RMS value of any of the two transverse components do not exceed the specified 10 μG (see Fig 24). The data in that plot has been obtained by measuring the fields every hour while keeping all the controllable parameters of the shield constant. Unfortunately, at the time of the measurement, I did not have available the means to monitor the magnetic field external to the magnetic shield, or at any other point inside the shield except on-axis.

Since degaussing is used to significantly reduce the internal magnetic field, it is of interest to observe whether degaussing could be used to periodically restore the initial low-field conditions even after shimming the fields. Unfortunately, it appears that degaussing leaves the shield in a state of a small but significant internal field. Since the shield is stable enough, I believe better performance is obtained by degaussing only at the beginning of measurements, then shimming the shield, and letting it stay that way.

It is true that just because the shield stayed in a low-field condition for three weeks

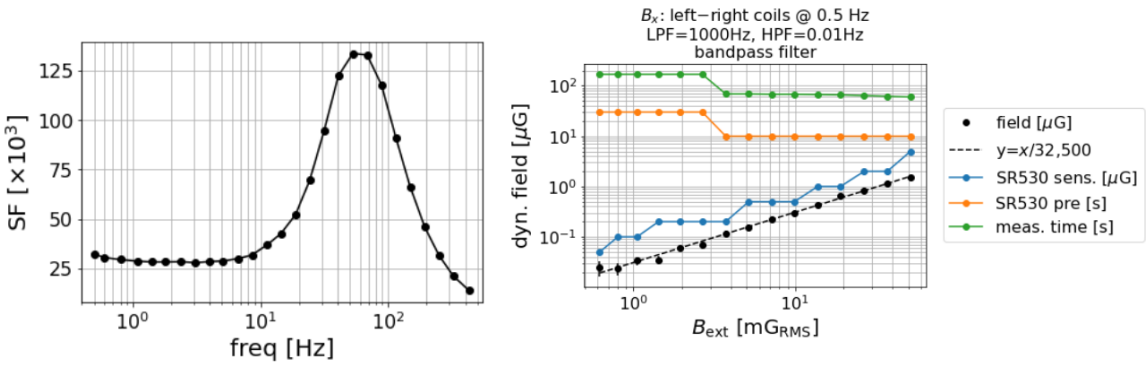

Figure 25: Dynamic shielding factor vs frequency and amplitude of the external field perturbation.

once, it is not certain it will always stay that way for as long. In other words, it would be useful to find a model for predicting how long the shield stays this way, or what the probability is that is has exceeded 10 μG. If we model the average internal magnetic field as a random walk, then we can associate a diffusion constant to this process and calculate the probabilities. Include the calculations for random walk. Of course, there is not physical reason why the growth in internal magnetic field should be a random walk; it would be of interest to compare other possible stochastic models for this behaviour.

In addition to the static internal field, it is also important to ensure the exteral magnetic field noise is sufficiently attenuated inside. My measurements of the dynamic shielding factor place it around 30,000 at DC (see Fig. 25), which should reduce the expected 1 mG variations in the lab field well below the level at which they would impact the molecular-beam measurements. The frequency dependence of the dynamic SF shown in that figure shows an increase in shielding as frequency increases, which makes sense in light of skin effect considerations. For mumetal of 1 mm thickness and relative permeability of around 50,000, a frequency of XX Hz would correspond to a skin depth comparable to the inner-layer thickness. (The decrease in shielding once the maximum is achieved is likely not physical and reflects some limitation of my measuring setup. I have, however, been so far unable to find out what causes this.)

The spatial dependence of the dynamic SF is as shown in Fig. 26. It will be noted that the pattern is asymmetric, as if one side of the shield works better than the other. That is also unlikely to be the case. Further work may establish whether insert possible

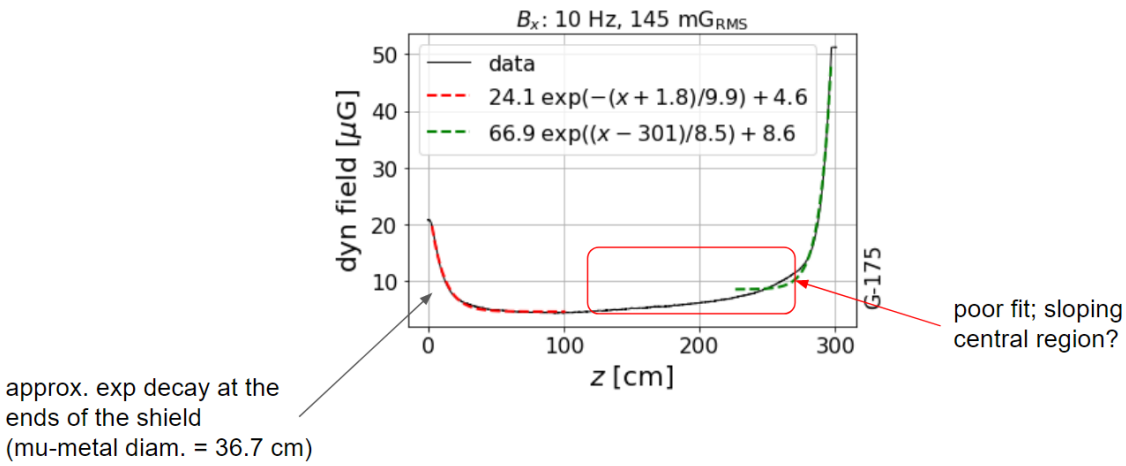

Figure 26: Spatial dependence of dynamic SF.

explanations from my to-do list.

While the dynamic shielding follows the approximate theoretical model, the static shielding does not. It is not clear why degaussing achieves such good shielding, nor is it known what limits it from working even better. It appears clear from a simple model that degaussing magnetizes the shield in such a way as to oppose the external magnetic fields, that being the configuration of lowest total energy.